BINF*6970 - Statistical Bioinformatics
Statistical Methods in Genetics Studies

Department of Mathematics and Statistics, University of Guelph

1. Basic Genetic Concepts and Terminologies

2. Linkage, Linkage Disequilibrium, Genetic Association Analysis

Basic Genetic Concepts and Terminologies

Some objectives of Genetic Studies

- Understand the biological system behind the complex inheritance system
- Identify the genetic causes of phenotypic variation
- Inheritance pattern in families.
- Complex disease gene mapping
- Animal bleeding improvement
- Better understand of evolution
- Evaluating the genetic risk of the occurrence of a disease.
- Drug development: finding genes responsible for a disease provide valuable insight of targeted pathway, personalized medicine.
- Increasing the fitness and the yield of crops under climate challenges, e.g., heat and drought climate

Some Branches in Genetic Studies

- Statistical Genetics
  - All types of analyses on inherited traits and genetic data.
  - Genes are referred as inheritance units from parents to offsprings.
  - Genes are units that could different from person to person.
  - Traits or phenotypes mean individual heritable characteristics, not observed at molecular level.
  - Methods/studies include: familial aggregation, segregation analysis, linkage analysis, association analysis, genome-wide association studies.
- Population Genetics
- Genetic Epidemiology
- Molecular Genetics

Some Branches in Genetic Studies

- Statistical Genetics
- Population Genetics
  - Concern with the genetic variation within and between populations, temporally and spatially.
  - Genetic selection due to environmental influences
  - In/out migration
  - Genetic drift in small populations
  - Mutations
  - Genetic differences in populations
  - Key principles in association analysis: Hardy-Weinberg equilibrium (HWE), linkage disequilibrium (LD), population stratification/substructure.
- Genetic Epidemiology
- Molecular Genetics

Some Branches in Genetic Studies

- Statistical Genetics
- Population Genetics
- Genetic Epidemiology
  - Interplay between genes, environments, and diseases
  - Use genetic data and metadata (e.g., geographic, demographic, spatial, temporal, racial, life style)
  - Concern variation in disease prevalence among populations,
  - More on observational studies, provide insight to the nature of disease.
- Molecular Genetics

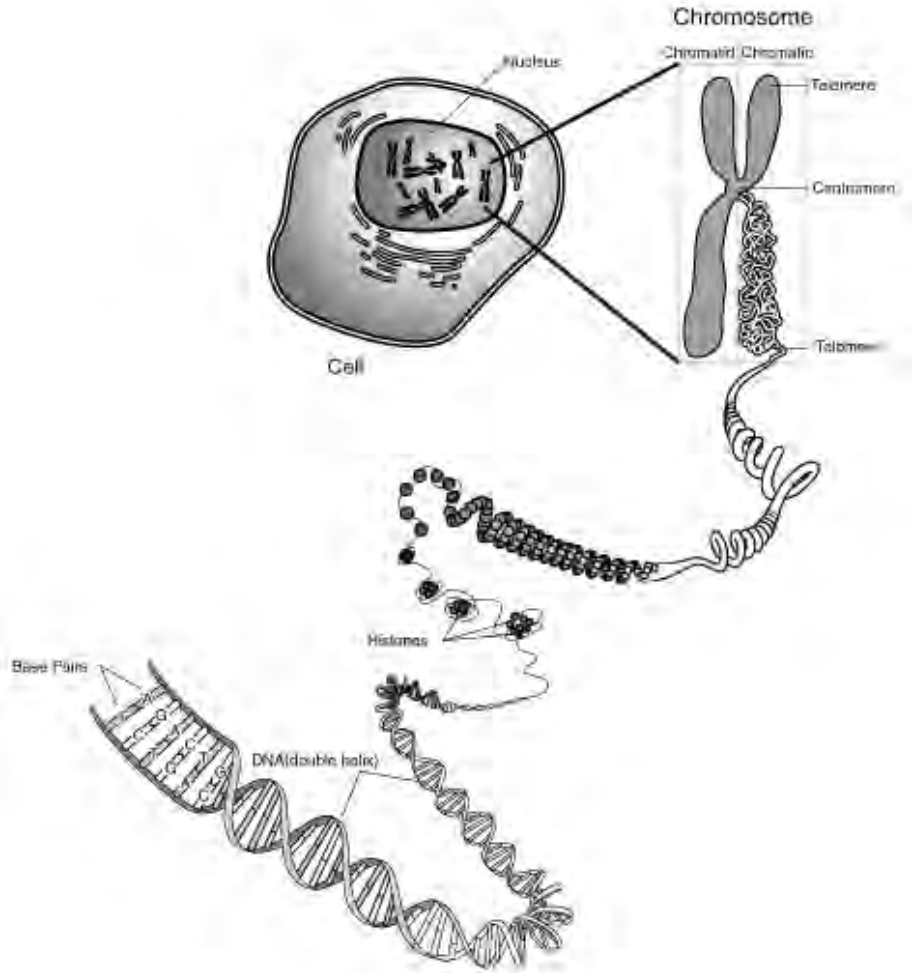
Molecular Genetics

Basic genetic terms:

genome, chromosome, DNA, genetic variants, locus, polymorphism, genetic marker, allele, genotype, homozygous, heterozygous, ...

Molecular Genetics

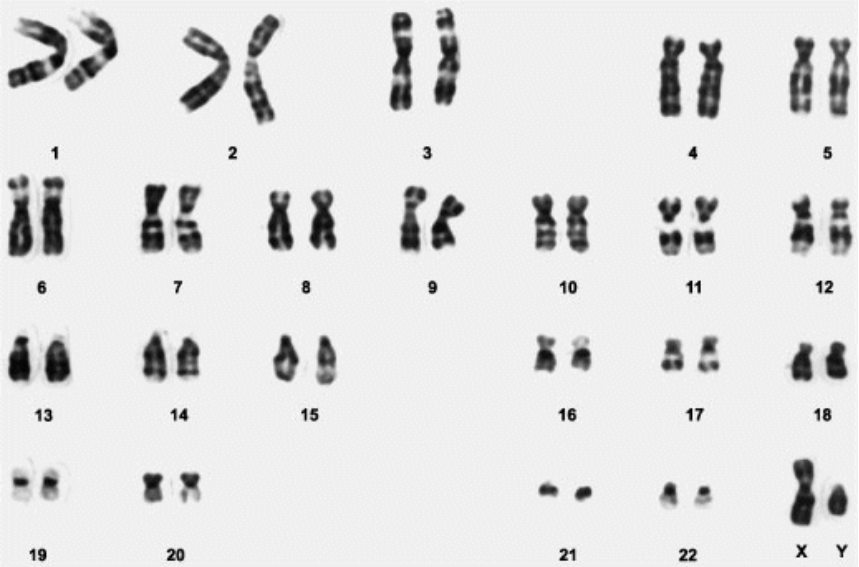
- Genome: all basic biological material that is transmitted from generation to generation.
- Chromosome: the heritable material packed into subunits in the nucleus of every cell.



Picture from http://www.accessexcellence.org/

Molecular Genetics

- Human genome consists of
  - 23 pairs of chromosomes: 22 pairs autosomes, one pair of sex-chromosomes
  - Two autosomes in a pair carry ALMOST identical genetic material are called homologous chromosomes (homologs).
  - Sex chromosomes: X and Y chromosomes
    Female: two X-chromosomes.
    Male: one X and one Y-chromosome.



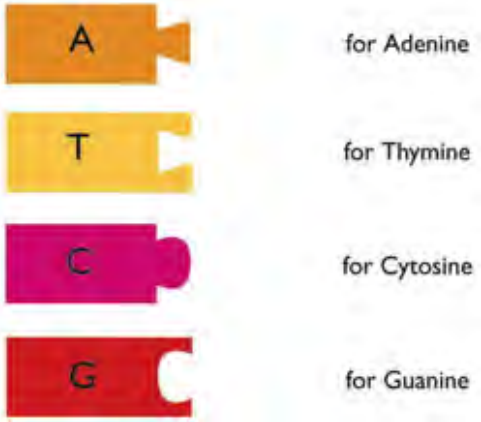
Picture from: www.genome.gov and www.answers.com

Molecular Genetics

- Centromere: a region near the middle of the chromosome.
  - Plays an important role in cell division and reproduction.
  - Specifies the chromosomal location.
  - Divides each chromosome into two arms: p (short arm, p for petit), and q (long arm, as q comes next to p alphabetically).
- Telomere: the two ends of a chromosome, hot spot of crossing over (location of exchange genetic material between two homologs).
- Each chromosome has two long strands, bounded to each other lengthwise with a double helix structure.

Deoxyribonuclei Acid (DNA)

- Basic biological material of inheritance
- Determines how proteins are manufactured in the body
- Composed of complementary base pairs
- Four distinct bases (A, C, T, G) which compose DNA in pairs



Picture from: info.cancerresearchuk.org.

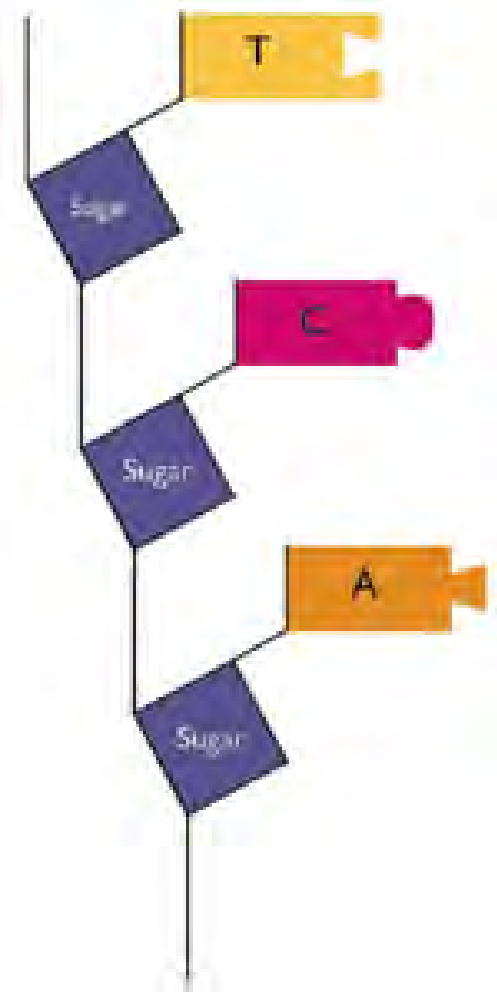
Deoxyribonuclei Acid (DNA)

- Complementary base pair: A-T, G-C.
- Information on one strand is sufficient:

A G T G C C A G A A

T T C A C G G T C T

A base pair (bp)



Picture from: info.cancerresearchuk.org

Deoxyribonuclei Acid (DNA)

- How DNA stores information: language based on 4 letters: ATGC.
  - Controlling centers for turning gene on and off.
  - Combinations of three letters from ATGC specify 20 amino acids - protein products.
  - Signaling the start of transcription - promoter.
  - Controlling the level of expression - transcription factor.
  - Signaling the end of protein.
- Human genome:
  - 3,000 millions (3×10^9) of DNA base pairs.
  - 99% identical across individuals, 1% make up genetic polymorphism.
  - Variation (genetic diversity across individuals) is the most interesting part. Responsible for individual differences, some are medical important.

Polymorphisms (Genetic Variants)

- A segment of DNA sequence or a single base that varies among individuals.
- Allele: different forms of a polymorphism.
- Causes of polymorphisms: mutations.
  - Introduce genetic diversity in populations.
  - Harmful mutation: change in DNA sequence cause errors in protein sequence, creating partially or completely non-functional proteins.
  - Beneficial mutations: positive effect from mutation that that new protein helps to have a better survivability.
  - Silent mutation: mutations has no effect on the function of the protein.
  - Only a small percentage of mutations have impact on health.

Polymorphisms (Genetic Variants)

- Types of mutations:
  - Substitution: a base is replaced by one of the other three bases.
  - Deletion: a block of one or more nucleotide pairs removed.
  - Insertion: addition of one or more nucleotide pairs.
  - Inversion: flip/reverse order of a segment of DNA.
  - Translocation: parts of two chromosome changes places.

Polymorphisms (Genetic Variants)

- Genes: protein coding sequences.
- Pseudogenes: Ancient genes, inactived through mutation.
- Promoters and transcription factors: sequences that control gene expression.
- Packaging sequences, "spacer" DNA, etc.
- Genetic markers.
- Locus (plural: loci): a location in a chromosome that is polymorphic.

Genes

- A segment of DNA that are responsible for making proteins.
- the beginning and end of a gene are signalled by specific short DNA sequences.
- $< 2\%$ of all DNA suggesting an estimate of 20,000-30,000 genes distributed through out the genome.
- Genes vary in size ranging from several thousands bp to millions bp.
- A gene consists of several coding segments (exon) separate by non-coding segments (introns).
- Intron (non-coding region): have no or unknown genetic effect on individuals.
  - Not junk, may play important roles in regulating the protein formation.
  - Important for gene mapping.

Genetic Markers

- Polymorphism with known location.
- With or with no (unknown) genetic effect.
- We characterize genetic markers by:
  - Number of alleles.
  - Allele frequency.
  - Heterozygosity.
- Common types of genetic markers:
  - Single Nucleotides polymorphisms (SNPs), mostly bi-allelic.
  - Microsatellites, multi-allelic.

Single Nucleotides polymorphisms (SNPs)

- Most common polymorphisms in human genome.
- Caused by single point mutation.
- Could be non functional.
- Could be functional relevant that occurs in a coding region.
  - Synonymous (silent) or non-synonymous: depending on whether or not it changes the amino acid sequence.
  - Missense: change one amino acid to another.
  - Nonsense: change an amino acid to a stop codon.
  - Frameshifts: insertion or deletion of one SNP completely change the subsequence of amino acids.
- Mostly bi-allelic: two allele types.

Microsatellite

- Short sequence of DNA repeated multiple times.
- Usually 3 to 30 different alleles.
- Highly polymorphic.
- No functional consequences.

SNP:

$$\cdots \text{CCGGTGATTATG} \begin{matrix} \text{A} \\ \text{G} \end{matrix} \text{ACCTACTATAA} \cdots$$

Microsatellite:

$$\cdots \text{AAGCTCTACACTACACTACACCTACTATAA} \cdots$$
$$\cdots \text{AAGCTCTACACTACACTACACTACACTACACTACACTACACCTACTATAA} \cdots$$

Genetic Data: Genotype

- Genotype: a pair of alleles that an individual constitutes at a given locus (chromosomal location).
- Example: ABO blood type.
  - Three alleles: A, B, and O.
  - Six genotypes: A/A, A/B, A/O, B/B, B/O, and O/O.
  - Homozygous (same allelic type): $A/A, B/B$, and O/O.
  - Heterozygous (different allelic type): $A/B, A/O$, and B/O.
- A marker with n alleles has:
  - n homozygous genotypes.
  - $\binom{n}{2} = n(n-1)/2$ heterozygous genotypes.

Genetic Data: Allele Frequencies

- For a given marker in an autosome, every individual has two alleles, one from each parent.
- Suppose the marker is bi-allelic that has two allele types: A and a.
- Allele frequencies are the proportions of each allele type in a given population, e.g., proportion of allele A and a, denoted by, p and $(1 - p)$ respectively.
- In a sample of size n from a given population, we want to estimate the $p = P(A)$.
- Usually, genetic data comes in genotypes, we count the genotype from the sample

Statistical Analysis: Estimate Allele Frequencies

- Given a genetic variant and a sample of n individuals, there are $2n$ alleles.
- Count the number of different genotypes in the sample:
  n_{AA} = number of individuals with homozygous AA
  n_{Aa} = number of individuals with heterozygous Aa
  n_{aa} = number of individuals with homozygous aa
  $n_{AA} + n_{Aa} + n_{aa} = n$.
- The number of allele A in the sample is given by

$$n_A = 2n_{AA} + n_{Aa},$$

the estimated allele frequency of A in the population is given by

$$\hat{p} = \frac{n_A}{2n} = \frac{2n_{AA} + n_{Aa}}{2n}$$

the estimated allele frequency of a is then $1 - \hat{p}$.

Population Genetics: Population Structure

- In genetics, population structure refers to the situation that a population can be divided into subpopulations.
- The allele distribution can be very different among different subpopulations.
- Based on the sample from a population with substructure, it might lead to bias estimate and/or inflated variance of estimate of the allele frequency.
- Common types of population structure:
  - population stratification
  - admixture
  - inbreeding

Population Stratification

- Simplest form of population substructure.
- A population can be divided into mutually exclusive strata
- Allele frequency is same within strata but varies among strata.
- Usually, different strata are represented by different ethnic groups.
- For example, residences in downtown Toronto can be classified into different ethnic groups: African, American, East Asian, European, and South Asian, according to the super-population classification from the 1000 Genomes Project.
  `http://www.internationalgenome.org/data-portal/population`

Statistical Analysis: Population Stratification

- Compare allele distributions of a given SNP among several population strata, for example, European (EU) vs South Asian (SA).
- Suppose we have two samples: n_1 and n_2, from EU and SA.
- EU sample: let $p_1 = P(\text{allele } A|EA)$, $\hat{p}_1 = \frac{n_{1,A}}{2n_1}$
  $2n_1$ alleles, $n_{1,A}$ allele A, and $n_{1,a} = 2n_1 - n_{1,A}$ allele a.
- SA sample: let $p_2 = P(\text{allele } A|SA)$, $\hat{p}_2 = \frac{n_{2,A}}{2n_2}$
  $2n_2$ alleles, $n_{2,A}$ allele A, and $n_{2,a} = 2n_2 - n_{2,A}$ allele a
- We want to determine whether the two populations: EU and SA, have the same allelic distribution at a given SNP.
- Perform a statistical hypothesis test based on the sampled data.

Statistical Hypothesis Test

- $H_0 : p_1 = p_2$ against $H_a : p1 \neq p_2$.
  We summarize the observed allele counts in a 2×2 contingency table

Population \ Allele	A	a	Row Total
European	$n_{1,A}$	$n_{1,a}$	$2n_1$
South Asian	$n_{2,A}$	$n_{2,a}$	$2n_2$
Column Total	n_A	n_a	$2n$

- Under H_0, $p_1 = p_2 = p$, we estimate p:
- The χ^2 test statistic is given by

$$X = \sum_{i,j=1}^{2} \frac{(O_{ij} - E_{ij})^2}{E_{ij}},$$

  Under H_0, $X \sim \chi^2_1$.
- At α significance level, we reject the H_0 if $X_{obs} > X^*$ (a threshold), or p-value $< \alpha$.

Statistical Hypothesis Test

- The O_{ij} is the observed count in the ith row and jth column of the table. The E_{ij} is the expected count for the (i, j) entry under the H_0.
- Under H_0:

$$E_{11} = 2n_1\hat{p}, \quad E_{12} = 2n_1(1-\hat{p})$$
$$E_{21} = 2n_2\hat{p}, \quad E_{22} = 2n_2(1-\hat{p})$$

- Compute X_{obs}

- Use R to find the $X^* = \chi^2_{1,1-\alpha}$.
  E.g. $\alpha = 0.05$, $1-\alpha = 0.95$, and $X^* = 3.841$.
  >qchisq(0.95, 1)
  [1] 3.841459

SNPs screening for population stratification

- Suppose you are about to screen for SNPs in the MC1R gene that have different allelic distribution among different populations.

  MC1R gene encodes the receptor protein for melanocyte-stimulating hormone (MSH). The encoded protein, a seven pass transmembrane G protein coupled receptor, controls melanogenesis. Two types of melanin exist: red pheomelanin and black eumelanin. Gene mutations that lead to a loss in function are associated with increased pheomelanin production, which leads to lighter skin and hair color. Eumelanin is photoprotective but pheomelanin may contribute to UV-induced skin damage by generating free radicals upon UV radiation. Binding of MSH to its receptor activates the receptor and stimulates eumelanin synthesis. This receptor is a major determining factor in sun sensitivity and is a genetic risk factor for melanoma and non-melanoma skin cancer. Over 30 variant alleles have been identified which correlate with skin and hair color, providing evidence that this gene is an important component in determining normal human pigment variation. (Source from: `https://www.ncbi.nlm.nih.gov/gene/4157`.)

SNPs in MC1R gene.

- Look up the information about MC1R gene in National Center for Biotechnology Information (NCBI), U.S. National Library of Medicine home page: `https://www.ncbi.nlm.nih.gov`
- MC1R in human is located on chromosome 16: 89984287..89987385bp.
- Use data slicer to extract SNPs from this genome region with your specified populations, e.g., FINN (Finland, Europe), BEB (Bengali, South Asia).
- For each SNP, perform the χ^2 test to test whether the two populations has the same allelic distribution.

Allelic distributions among several populations

- Suppose you are interested in comparing the allelic distribution of a SNP among several, sys K, populations. You can analyze that data in a $K \times 2$ contingency table and perform an overall test .
- Comparisons among 6 subpopulations within East Asia (EA).

Population \ Allele	A	a	Row Total
JPT, Japanese in Tokyo	$n_{1,A}$	$n_{1,a}$	$2n_1$
CDX, Chinese Dai in Xishuangbanna	$n_{2,A}$	$n_{2,a}$	$2n_2$
CHB, Han Chinese in Beijing	$n_{3,A}$	$n_{3,a}$	$2n_3$
CHS, Han Chinese South	$n_{4,A}$	$n_{4,a}$	$2n_4$
KHV, Kinh in Ho Chi Minh City	$n_{5,A}$	$n_{5,a}$	$2n_5$
CHD, Chinese in Denver,US	$n_{6,A}$	$n_{6,a}$	$2n_6$
Column Total	n_A	n_a	$2n$

Allelic distributions among several populations

- $H_0 : p_1 = p_2 = ... = p_K$ against H_a :not all p's are equal.
- Under H_0, $p_1 = p_2 = ... = p_K = p$, we estimate p:
- The χ^2 test statistic is given by

$$X = \sum_{i=1}^{K} \sum_{j=1}^{2} \frac{(O_{ij} - E_{ij})^2}{E_{ij}},$$

  Under H_0, $X \sim \chi^2_{K-1}$.
- At α significance level, we reject the H_0 if $X_{obs} > \chi^2_{k-1,1-\alpha}$, or p-value $< \alpha$.

Population Admixture

- A situation where individuals in a population have a mixture of different genetic ancestries due to the mixing of two or more populations at a previous time point.
- Results from a migration of one or more population groups from their original regions to a new geographical region with a previously settled population.
- Allele frequencies of the admixture population might be different from the original ancestral populations if the involved population have different allele frequencies.

Population Admixture

Example: Native Americans in the Pima and Papago tribes have different degrees of American Indian and European Hispanic ancestry.

- The below table summarizes the frequency of $GM3;5;13;14$ allele on the human immunoglobulin G gene, the percentage with diabetes for the adults in the population, stratified by the number of great grandparents with Indian heritage.
- Higher degree of Indian ancestry, the lower the allele frequency, higher incidence of diabetes.
- Can we draw a conclusion about the protective effect of the $GM3;5;13;14$ allele on the development of diabetes?
- Be careful about the spurious association between disease and marker.

Great grandparents	$GM3;5;13;14$ frequency	%Diabetes
0	0.658	18.5%
4	0.421	28.5%
8	0.016	39.2%

Adapted from Knowler et al. (1988)

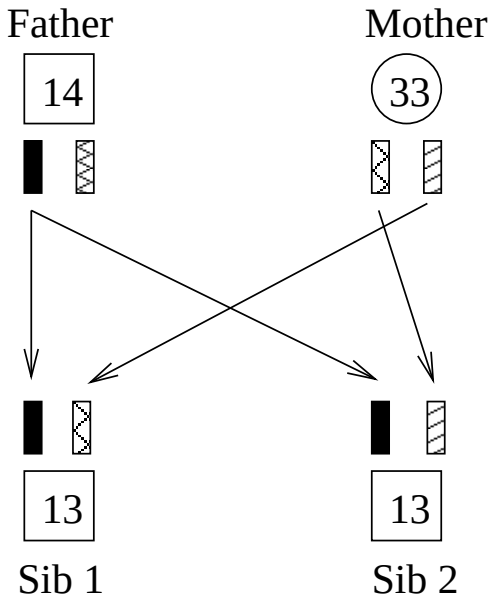
Population Inbreeding

- A situation when there is a preference for mating among relatives in a population.
- Common in isolated/founder population and breeding animal populations.
- Inbreeding coefficient (Φ) : probability that an individual inherits two alleles that are identical by descent (IBD).
- In an outbred population, or a random mating populations, the chance of any two mating parents have a common ancestor allele is very low, i.e, $\Phi \approx 0$.
- At the extreme, self-breeding plants have $\Phi = 1$.
- The Φ for an individual reflects the relationship between the parents of the individual.

Identical by descent (IBD)

- IBD: at a give locus, the two alleles are inherited from the same origin (common ancestor), we say they share one allele IBD.
- E.g., the IBD sharing is:

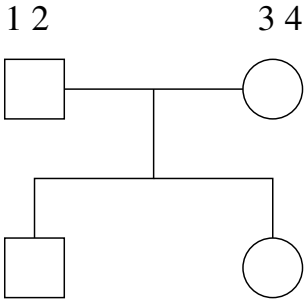
  0 for two parents,
  1 for parent and offspring,
  1 for two sibs.



- A measure of similarity of inheritance mode and similarity of genetic material among relatives.

Statistical analysis: IBD distribution

- Under the Mendelian inheritance model, the probability distribution of IBD sharing can be inferred for a pair of relatives.
- E.g., for a pair of siblings,



Sibling's genotype can be 13, 14, 23 or 24 with equal probability.

- Thus, we can have 16 possible combination of genotypes for a pair of siblings.

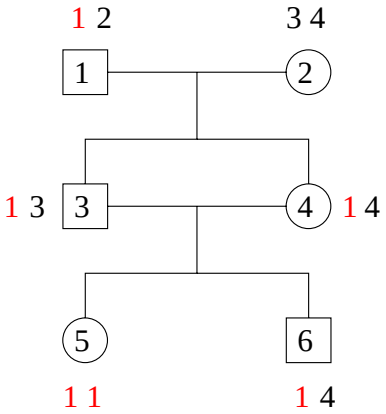
genotype	IBD	gentoype	IBD
13, 24	0	14, 24	1
13, 23	1	14, 23	0
13, 14	1	14, 14	2
13, 13	2	14, 13	1
23, 24	1	24, 24	2
⋮	⋮	⋮	⋮

- The IBD distribution for a pair of individual can be denoted by $\pi = (\pi_0, \pi_1, \pi_2)$, where p_i =P(two individuals share i alleles IBD).
- For a pair of siblings: $\pi = (\pi_0, \pi_1, \pi_2) = (1/4, 1/2, 1/4)$.
- The IBD distribution for a pair of individuals depends on their relationship.

Relationship	IBD distribution
unrelated	$\pi = (1,\ 0,\ 0)$
parent-child	$\pi = (0,\ 1,\ 0)$
monozygous twin	$\pi = (0,\ 0,\ 1)$
full sibs	$\pi = (1/4,\ 1/2,\ 1/4)$
half sibs, grandparent, aunt	$\pi = (1/2,\ 1/2,\ 0)$
first cousin	$\pi = (3/4,\ 1/4,\ 0\ )$

Kinship-Inbreeding coefficients

- Suppose a family has 6 members: individual 1, 2, 3, ..., 6 with their genotypes information available at a given marker. E.g., individual 1 has genotype 12.



ibd(1)=0, ibd(1,2)=0

ibd(3)=0, ibd(3,4)= 1

ibd(5)=1, ibd (6)=0, ibd (5,6)=1

- Inbreeding coefficient Φ_i: the probability that the two alleles of individual i are ibd.
- Kinship coefficient Φ_{ij}: the probability that an allele selected randomly from individual i and an allele selected randomly from individual j are ibd.

Kinship-Inbreeding coefficients matrix: K

$$\mathbf{K} = \begin{pmatrix} 1+\Phi_1 & 2\Phi_{12} & \cdots & 2\Phi_{16} \\ 2\Phi_{12} & 1+\Phi_2 & \cdots & 2\Phi_{26} \\ \vdots & \cdots & \ddots & \vdots \\ 2\Phi_{16} & 2\Phi_{26} & \cdots & 1+\Phi_6 \end{pmatrix}$$

Population Genetics: Hardy-Weinberg Equilibrium (HWE)

Formulated in 1908 independently by

- Hardy G H, English mathematician.
- Weinberg W, German physician.

In a given population

- Allele frequency: proportion of allele A and a, denoted by, p and $(1-p)$ respectively.
- Genotype frequency: proportion of genotype AA, Aa, and aa, denoted by,p_{AA}, p_{Aa}, p_{aa},respectively..
- HWE: specifies a particular relationship between allele frequency and genotype frequency,

$$p_{AA} = p^2, \quad p_{Aa} = 2p(1-p), \quad p_{aa} = (1-p)^2.$$

Hardy-Weinberg Equilibrium (HWE)

- Many assumptions are required for the HWE to be held:
  - random mating, no inbreeding
  - infinite or large population size
  - discrete generations
  - no disturbing forces, e.g.,migration, mutation, selection (alleles do not confer a selective advantage or disadvantage in reproduction).
- In reality, these assumptions are hard to meet in any population. However, when population size is very large, events that violate the assumptions become rare or negligible.
- If a population is in HWE, then
  - two alleles in a genotype are independently drawn from the same distribution.
  - the genotype frequencies in the population depend only on the allele frequencies.

Statistical Analysis: Test of HWE

- Importance of HWE:
  - Many methods and approaches in genetic studies require the assumption of HWE.
  - Some methods may not be robust to the HWE assumption.
  - Need to test HWE before applying methods.
- Checking the satisfactory of HWE is the first step in some genetic studies.
- Deviation of HWE can be due to but not limited to:
  - genotyping error,
  - unexpected population structure,
  - existence of highly homozygous regions
  - association with traits in case-control studies...

Pearson χ^2 goodness-of-fit test of HWE

H_0 : HWE holds against H_a: HWE does not hold

- Given a sample of size n from a population and a SNP, we compare the observed genotype frequencies with the expected frequencies calculated under the assumption of HWE.

	Genotype			
	AA	Aa	aa	
Observed frequency O	n_{AA}	n_{Aa}	n_{aa}	n
Expected Frequency E	np^2	$2np(1-p)$	$n(1-p)^2$	n

$$X = \sum_{\text{genotype } i} \frac{(O_i - E_i)^2}{E_i}.$$

Pearson χ^2 goodness-of-fit test of HWE

- Because p is most likely not given, estimate p using the observed data:
$$\hat{p} = \frac{2 * n_{AA} + n_{Aa}}{2 * n}.$$
- Compute the Expected frequency:
$$E_{AA} = n\hat{p}^2, \quad E_{Aa} = 2\hat{p} * (1 - \hat{p}), \quad E_{aa} = n * (1 - \hat{p})^2.$$
- Under the H_0, X follows the χ^2_1 distribution.
- Compute the observed X statistic: X_{obs}.
- At the α level, reject the H_0, i.e. HWE holds if $X_{obs} > X^* = \chi^2_{1,1-\alpha}$, or, otherwise, do not reject the H_0.
- Alternatively, reject the H_0 if the p-value $< \alpha$.

Linkage, Linkage Disequilibrium, Genetic Association Analysis

Mendel's Law of Inheritance

- Gregor Mendel (1822-1884), who is known as the "father of modern genetics", stated two famous inheritance laws: law of segregation and law of independent assortment.
- (First) Law of segregation: during gamete formation, the alleles for each gene segregate from each other so that each gamete carries only one allele for each gene.
- (Second) Law of independent assortment: Genes for different traits can segregate independently during the formation of gametes. In other word, the segregations at two loci are independent.
- Mendel's second law is partially true, it is true only when the two loci are on different chromosomes, i.e., the two loci are not linked.

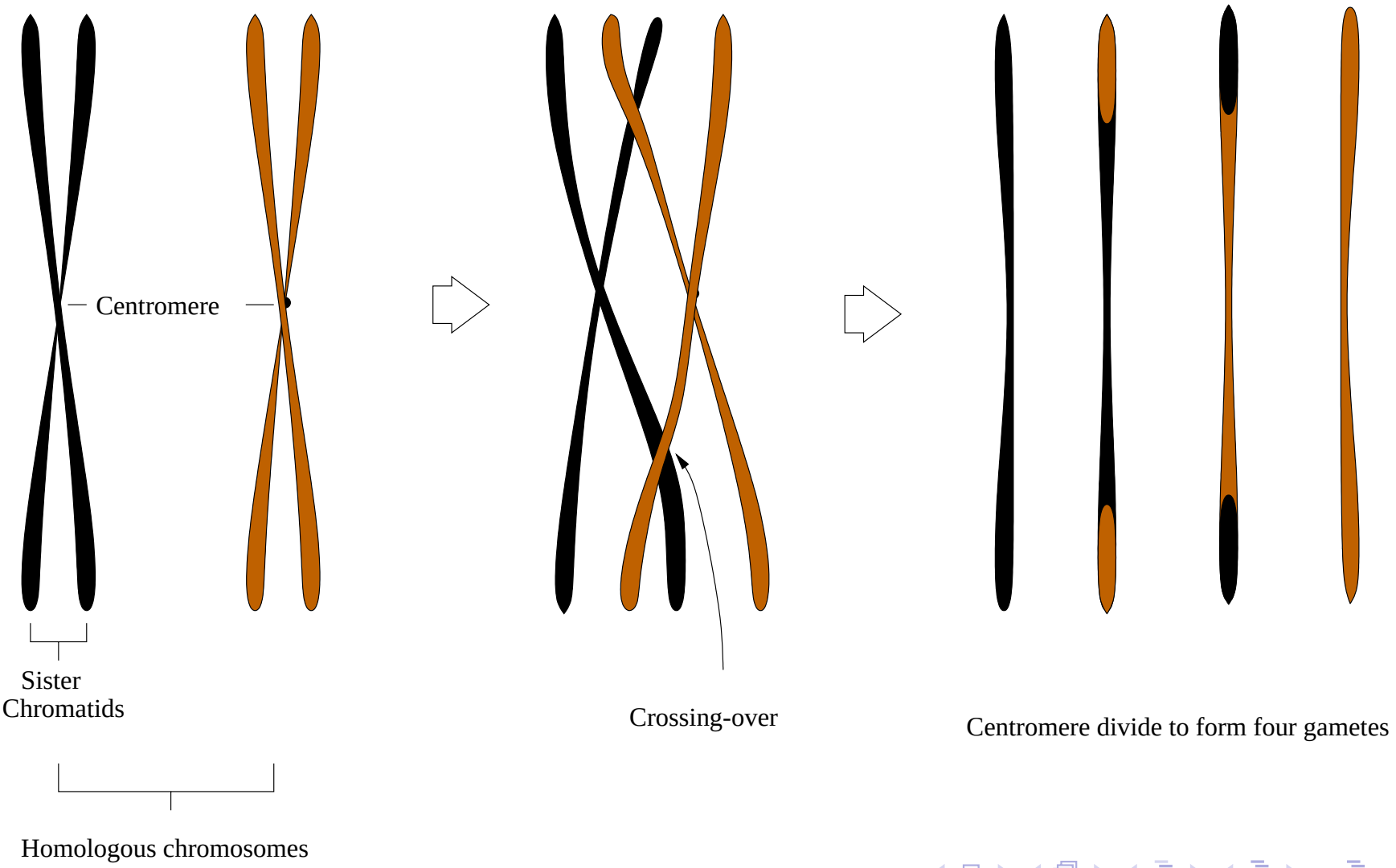
Morgan's Fruit Flies studies

- Thomas Hunt Morgan (1866-1945), an American evolutionary biologist, geneticist, and embryologist who won the Nobel Prize in Physiology or Medicine in 1933 for discovering the role of chromosomes ply in heredity.
- In his fruit flies study, he discovered that the segregation of the two traits (genes/loci) are not independent when the two loci are on the same chromosome arm.
- This can be explained by the crossover event between homologous chromosomes during the meiosis.

Meiosis, Crossover, and Recombiation

- Meiosis: a cell division process (reproduction process) during which an egg or a sperm cell is formed.
- Crossover and recombination: an important process during the meiosis.
  - each pair of homologous chromosomes pair up and duplicate to produce four strands of chromosomes.
  - Homologous chromosomes crossover to exchange genetic materials.
  - Then they separate and form egg or sperm within which has only one strand.
  - The transmitted chromosome strand could be a mixture of the paternal and maternal chromosome depending on if there is a crossover or not.
  - If the strand was involved in an odd number of crossover between two loci, there is a recombination between the two loci.
  - Crossovers are independent from different homologous chromosome pairs.
- A group of alleles along the single strand forms a haplotype.

Crossovers on a pair of chromosomes during the meiosis



Genetic Map

- If two loci are far apart, it is more likely to have a crossover in between.
- The expected number of crossover events between two loci is used to measure the distance between them.
- Unit: Morgan or centiMorgan (cM), 1Morgan = 100cM.
- If two loci are 1cM apart, we expected to see about 1 crossover occur between the two loci in 100 gametes.
- Genetic distance defined on one chromatid strand, or within an arm.
- Human genome: sum of about 30 Morgan.
- Chromosome 1: about 2 Morgan. (There are about 4 crossovers during the meiosis.)
- Chromosome 22: about 0.5 Morgan. (There is about 1 crossover during the meiosis.)

Genetic Map

- Genetic distance $\neq$ physical distance.
- Physical distance: number of base pairs.
- Human genome: 3×10^9 bp.
- Relationship between Genetic distance and physical distance is not linear.
- Crossover rate varies along a chromosome (hot spot and desert) and among chromosomes.
- In general, 1 cM $\approx$ 1Mb (1 million bp, 1000kb).
- In practice, crossovers cannot be observed directly in human data.

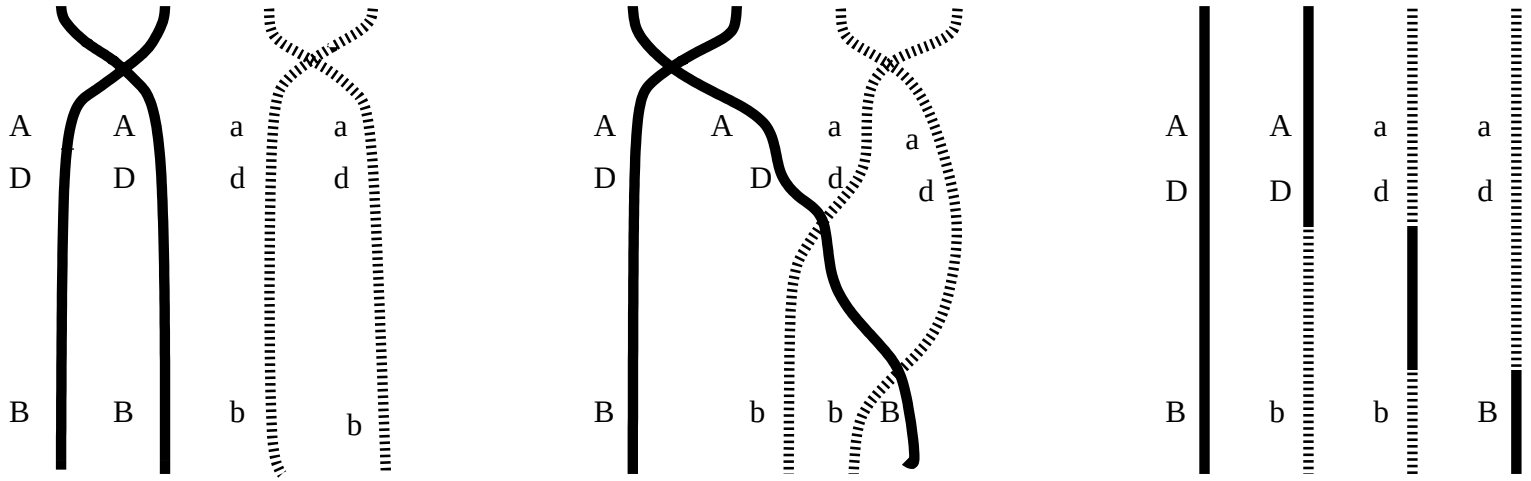
Human Genome: Chromosome Length Genetic Distance and Physical Distance

Chromosome	Physical Length (Mb)	Genetic Length, (cM)		Chromosome	Physical Length (Mb)	Genetic Length, (cM)	
		Male	Female			Male	Female
1	263	221	376	13	144	107	157
2	255	193	297	14	109	106	151
3	214	186	289	15	106	84	149
4	203	157	274	16	98	110	152
5	194	149	267	17	92	108	152
6	183	142	222	18	85	111	149
7	171	144	244	19	67	113	121
8	155	135	226	20	72	104	120
9	145	130	176	21	50	66	77
10	144	144	192	22	56	78	89
11	144	125	189	X	160	-	193
12	143	136	232	$\sum^1$	3093	2849	4301

1. Sum over all autosomes

Source from Morton (1991), PNAS USA, 88:7474-7476

Disease gene mapping



- D/d is a disease gene, A/a and B/b are two markers.
- Among the four gametes, there are recombinant haplotypes and parental haplotypes:
  - there is no crossover and so no recombinant between disease gene D and marker B in the 1st gamete.
  - there is one crossover and so a recombinant between disease gene D and marker B in the 2nd and 4th gametes.
  - there is 2 crossovers and so no recombinant between disease gene D and marker B in the 3rd gamete.
  - Overall, disease gene D and marker B, there are 2 out of 4 recombinant haplotypes, and 2 out 4 parental haplotypes.

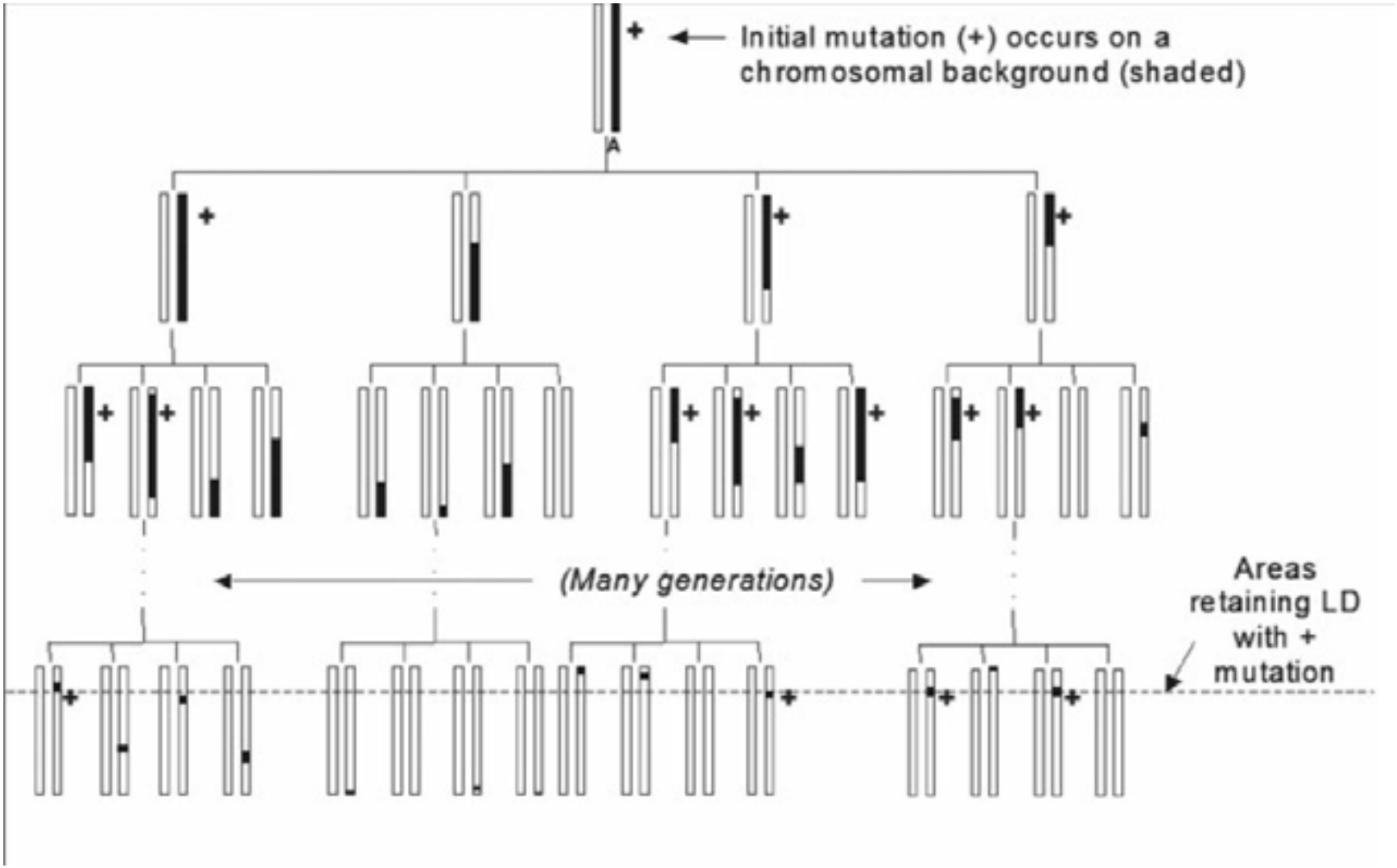
Disease gene mapping

- Between disease gene D and marker A, there is no crossover and so no recombinant among four gametes, all gametes have a parental haplotype.
- The number of crossovers between two loci in a gamete determine the recombination event:
  - event number of crossovers, no recombinant,
  - odd number of crossover, recombinant.
- The further distance between two loci, the higher chance of having crossovers and higher chance of getting a recombinant gamete to the offspring.
- Recombination fraction θ between two loci: the probability that there is recombination between the two loci during a single meiosis.

Disease gene mapping

- Property of recombination fraction θ:
  - θ has upper bound of $\frac{1}{2}$, i.e., $0 \leq \theta \leq \frac{1}{2}$.
  - $\theta = 0$, complete or perfect linkage.
  - $\theta = \frac{1}{2}$, unlinked. That is, alleles at the two loci segregate independently - mendel's law of independent assortment.
  - $\theta < \frac{1}{2}$: linkage.
  - θ is not additive.
  - θ =P(wo alleles at two loci are from different strands)
    =P(a recombinant haplotype).
- Unlike crossover event, recombination events may be observed through study design.
- Crossing-over during meiosis allows to exchange genetic material between the paternal and maternal haplotypes and generate offsprings that have mixed genetic material from both parents.
- Using high throughput information of genetic variants, e.g., SNPs, disease gene loci, or other trait loci of interest can be identified through linkage analysis or association analysis.

Linkage and Linkage disequilibrium



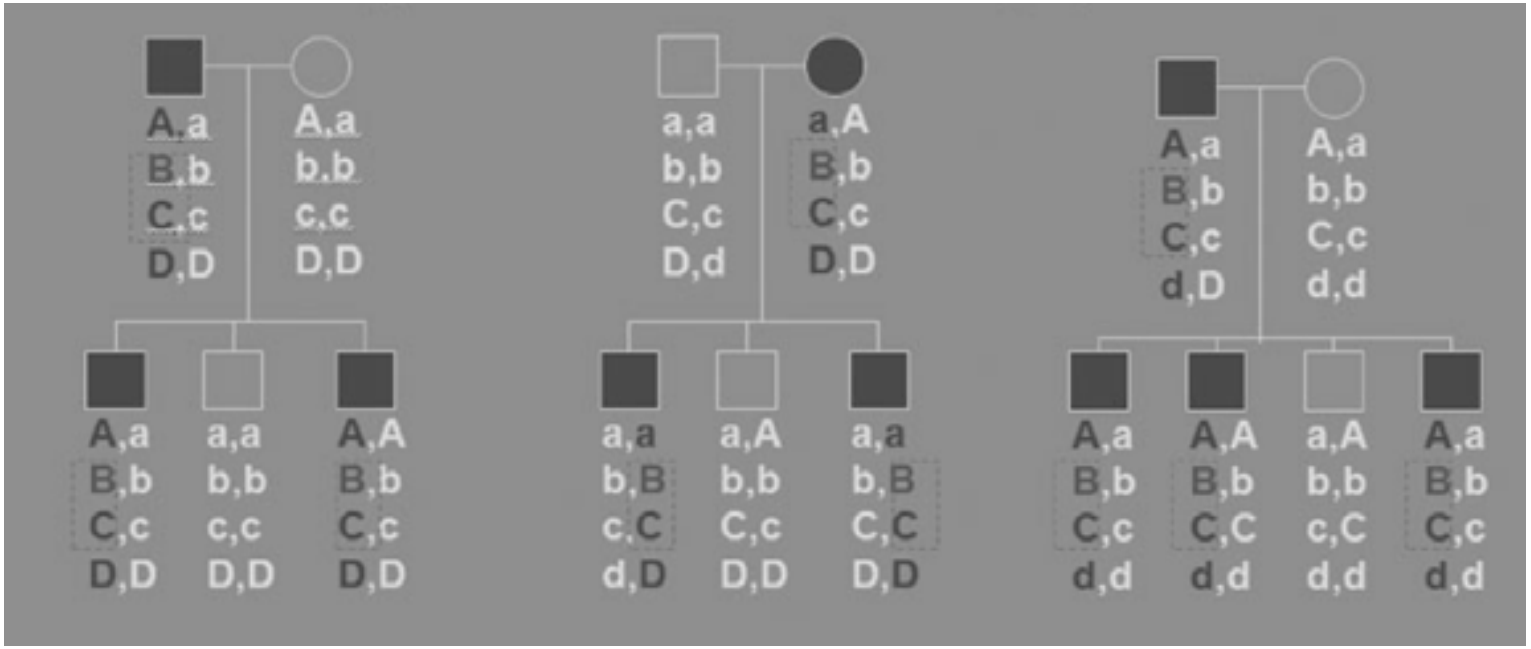
Source from Laird and Lange (2011), The Fundamentals of Modern Statistical Genetics, Springer Series of Statistics for Biology and Health, Editors: Gail, Tsiatis, Krickeberg, and Wong.

Linkage Analysis

- Linage analysis is performed using data collected from families.
- Investigate the co-seggregation pattern between genetic variants and a trait or disease affection status from a generation to next generation in order to localize the disease/trait gene(s).
- Formally, test the $H_0 : \theta = \frac{1}{2}$ vs $H_a : \theta < \frac{1}{2}$.

Linkage Analysis and Association Analysis

Figure 1: Linkage and Association



Source from Laird and Lange (2011), The Fundamentals of Modern Statistical Genetics, Springer Series of Statistics for Biology and Health, Editors: Gail, Tsiatis, Krickeberg, and Wong.

Linkage Analysis

- Suppose there are three markers with alleles A,a, and C,c, and D, d.
- A disease gene has disease causing allele B and wild type allele b.
- The two haplotypes of each individual are separated by ",".
- It is observed that all three markers alleles are transmitted together with the disease allele as a parental haplotype from the affected parent to the affected offspring.
- All three markers are linked to the disease locus.
- Suppose the disease locus B is unknown, through the co-segregation pattern between the markers and disease trait, it is location can be narrowed down to the neighbourhood of the three markers.

Linkage Disequilibrium, Association

Refer to Figure 1,

- Different alleles: A or a, D or d are transmitted with the disease causing allele B among these families.
- Only allele C is transmitted with the allele B in all families.
- The allele C is associated with the allele B in the population level.
- The association of two alleles at two loci is referred to as linkage disequilibrium (LD).
- The associated alleles tends to transmit together as a parental haplotype from generation to generation.
- The absence of association between two loci is called linkage equilibrium (LE).
- Genetic associate analysis is referred to as the identification of a genetic marker associated with a disease/trait of interest.

LD/Association

- We assume that HWE holds in a given population.
- LD/association between two loci refers to the lack of independence (allelic association) of alleles at loci on a haplotype randomly selected from a population.
- Two particular alleles at two loci are more likely to be on the same chromosome than what is expected by random mating.
- Suppose that allele A_i at locu A has frequency p_{A_i} and allele B_j at locus B has frequency p_{B_j}.
- Let $P(A_iB_j)$ be the frequency of haplotype A_iB_j in the population.
- Mathematically, two loci A and B are in LD or associated if

$$P(A_iB_j) \neq p_{A_i}p_{B_j}.$$

LD Measure

- Suppose locus A has allele A and a and locus B has allele B and b.
- $D = p_{ab} - p_a p_b$, where p_{ab} is the frequency of haplotype ab and p_a and p_b are the frequencies of allele a and allele b.

  D is called the disequilibrium parameter.

  $D = 0$: linkage equilibrium.
  $D > 0$: positive disequilibrium or association. Or, otherwise, negative disequilibrium or association.
- Since each frequency takes values in between 0 and 1, D can only take values in a range of

  $D_{max} = min(p_a(1 - p_b), (1 - pa)p_b)$,
  $D_{min} = max(-p_a p_b, -(1 - p_a)(1 - p_b))$.
  A standardized disequilibrium:

  $$D' = \frac{D}{D_{max}} \text{ or } D' = \frac{D}{D_{min}},$$

  depending on whether D is positive or negative.

Association Study

- Population based association studies: case-control study, quantitative trait study.
- Family based methods
  - Transmission disequilibrium test (TDT) method.
  - Family-based association test (FBAT).
  - Case-control association test with related samples.
  - Generalized Quasi-likelihood scoring method.
- Spurious association

Association Study

- Suppose a SNP has two alleles: A and a, and a disease locus has two alleles: D and d.
- We assume they are in complete LD, that is, allele a and mutant D ar always on a haplotype.
- Among haplotypes that have mutant D, what is the allele frequency of allele a?

$$P(a|aD \text{ or } AD) = \frac{P(aD)}{P(aD) + P(AD)} = \frac{P(aD)}{P(aD)} = 1.$$

- Among haplotypes that have normal allele d, what is the allele frequency of allele a?

$$P(a|ad \text{ or } Ad) = \frac{P(ad)}{P(ad) + P(Ad)} = \frac{0}{f_d} = 0$$

- Comparing the group of disease affected individuals (case group) with the group of disease free individuals (control group), the frequency of allele a will be higher in the case group than in control group.

Association Study

- Case-control association analysis: examine if the allelic distribution of a given mark is different between the case group and control group.
- Suppose you have two samples: one sample of size n_1 cases and another sample of size n_2 controls.
- A SNP under investigation has two alleles: 1 and 2
- The data is summarized in the table

	SNP		
group	allele 1	allele 2	Total
case	$n_{1,1}$	$n_{1,2}$	$2n_1$
control	$n_{2,1}$	$n_{2,3}$	$2n_2$
	$n_{.1}$	$n_{.2}$	$2n$

Association Study

- Case-control association analysis under HWE: examine if the allelic distribution of a given mark is different between the case group and control group.
- Let p_1 and p_2 be the allele frequencies of allele 1 in the case population and the control population.
- We perform a χ^2 test: $H_0 : p_1 = p_2 = p$, against the $H_a : p_1 \neq p_2$.
- Again, the χ^2 statistic is given as

$$X = \sum_{i,j=1}^{2} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}.$$

  Under $H_0, X \sim \chi^2$.
- Test whether HWE holds, using control group.
- Be careful about the spurious association: the association between a trait (disease) and a marker that is not linked to any trait causal locus.
- Possible causes: population stratification.

Association Test When HWE Does Not Hold.

- Recall that the HWE specifies a particular relationship between allele frequency and genotype frequency,

$$p_{AA} = p^2, \quad p_{Aa} = 2p(1-p), \quad p_{aa} = (1-p)^2$$

  where p and $(1-p)$ are the frequencies of allele A and a and (p_{AA}, p_{Aa}, p_{aa}) are frequencies genotype AA, Aa, and aa, respectively, in a given population
- When HWE doesn't hold, we use the Armitage trend test.
- Suppose the data is summarized in terms of genotype frequencies in the table

	SNP Genotype			
group	AA	Aa	aa	Total
case	$n_{1,0}$	$n_{1,1}$	$n_{1,2}$	n_1
control	$n_{2,0}$	$n_{2,1}$	$n_{2,2}$	n_2
	$n_{.0}$	$n_{.1}$	$n_{.2}$	n

Armitage Trend Test.

- Let Y be an binary variable that $Y_i = 1$ if the ith subject is case and 2 if a control.
- Let $X = 0, 1, 2$ represent the three genotypes of a SNP, or the count of allele a of a SNP.
- Two populations: case and control populations, we have two samples: $X_1 = (X_{11}, ...X_{1,n_1})$ and $X_2 = (X_{21}, ...X_{2,n_2})$ of genotypes, from the two populations, respectively.
- Let μ_1 and μ_2 be the means of X from the case and control samples respectively.
- If the SNP is not associated with the disease/trait, $\mu_1 = \mu_2$.
- Test $H_0 : \mu_1 = \mu_2$ agains $H_a : \mu_1 \neq \mu_2$.
- Use the two-sample t-test.

Armitage Trend Test.

- $X_{1i} \sim Binomial(2, p_1)$, where p_1 is frequency of allele a in case population,
  $X_{2i} \sim Binomial(2, p_2)$, where p_2 is frequency of allele a in control population.
- $\mu_1 = E(X_{1i}) = 2p_1$, $\mu_2 = E(X_{2i}) = 2p_2$,
- $H_0 : \mu_1 = \mu_2$ agains $H_a : \mu_1 \neq \mu_2$.
- $\hat{\mu}_1 = \bar{X}_1$ and $\hat{\mu}_2 = \bar{X}_2$, where
  $\bar{X}_1 = \frac{2n_{1,2}+n_{1,1}}{n_1}, \bar{X}_2 = \frac{2n_{2,2}+n_{2,1}}{n_2}$, $\hat{\mu}_1 - \hat{\mu}_2 = \bar{X}_1 - \bar{X}_2$.
  Assuming both populations have equal variances:
  $var(X_{1i}) = var(X_{2i})$, $var(\bar{X}_1 - \bar{X}_2) = var(X)(\frac{1}{n_1} + \frac{1}{n_2})$.
  Estimate $var(X)$ by the pooled sample variance

$$\hat{var}(X) = \frac{\sum_1^{n_1}(X_{1i} - \bar{X}_1)^2 + \sum_1^{n_2}(X_{2i} - \bar{X}_2)^2}{n-2}$$

Armitage Trend Test.

- Under $H_0 : \mu_1 = \mu_2$ is equivalently $\mu_1 - \mu_2 = 0$,
- Without assuming HWE, we use $t = \frac{\hat{\mu}_1 - \hat{\mu}_2}{SE(\hat{\mu}_1 - \hat{\mu}_2)} \sim t_{n_1+n_2-2}$ under H_0, where

$$SE(\hat{\mu}_1 - \hat{\mu}_2) = \sqrt{\hat{var}(\hat{\mu}_1 - \hat{\mu}_2)} = \sqrt{\hat{var}(X)(\frac{1}{n_1} + \frac{1}{n_2})}$$

- Compute the t_{obs} and obtain the p-value$=2P(t_{n_1+n_2-2} > |t_{obs}|)$, reject H_0 if p-value< 0.05.
- For example, we have data summarized in the table

	SNP Genotype			
group	AA	Aa	aa	Total
case	66	43	4	113
control	99	15	0	114
	165	58	4	227

Armitage Trend Test

In R, we use the function t.test( ).

```
> x=c(rep(0, 66), rep(1, 43), rep(2,4))
> y=c(rep(0, 99), rep(1,15))
> model=t.test(x,y,var.equal = TRUE)
> model

Two Sample t-test

data:  x and y
t = 5.1611, df = 225, p-value = 5.389e-07
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 0.1976651 0.4418319
sample estimates:
mean of x mean of y
0.4513274 0.1315789
> model$p.value
[1] 5.389331e-07
```

Mode of Inheritance

- Association test concerning different modes of inheritance: codominant, dominant, recessive, and additive.

	SNP Genotype			
group	AA	Aa	aa	Total
case	$n_{1,0}$	$n_{1,1}$	$n_{1,2}$	n_1
control	$n_{2,0}$	$n_{2,1}$	$n_{2,2}$	n_2
	$n_{.0}$	$n_{.1}$	$n_{.2}$	n

- The codominant test is a genotype test:
  $H_0 : P(\text{case}|AA) = P(\text{case}|Aa) = P(\text{case}|aa)$

  H_a : at least one inequality holds.
  - Under H_0: $P(\text{case}|AA) = P(\text{case}|Aa) = P(\text{case}|aa) = p_{case}$, we estimate p_{case} by $\hat{p}_{case} = \frac{n_1}{n}$
  - We use the χ^2 test

$$X = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}.$$

Under H_0, $X \sim \chi^2_2$.

Mode of Inheritance

- Test for dominant and recessive modes of inheritance.
- Summarize the 2×3 table to a 2×2 table. For example, a dominant model of allele a (a recessive protective allele A).

	SNP Genotype		
group	AA	Aa or aa	Total
case	$n_{1,0}$	$n_{1,1} + n_{1,2}$	n_1
control	$n_{2,0}$	$n_{2,1} + n_{2,2}$	n_2
	$n_{.0}$	$n_{.1} + n_{.2}$	n

- $H_0 : P(\text{case}|AA) = P(\text{case}|Aa) = P(\text{case}|Aa \text{ or } aa)$

  H_a : at least one inequality holds.
  - Under H_0: $P(\text{case}|AA) = P(\text{case}|Aa \text{ or } aa) = p_{case}$, we estimate p_{case} by $\hat{p}_{case} = \frac{n_1}{n}$
  - We use the χ^2 test

$$X = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}.$$

  Under H_0, $X \sim \chi^2_1$.

Mode of Inheritance

- The allelic test and the trend test are for testing the genetic association under the additive model of inheritance.
- Allelic test assumes that HWE holds.
- The trend test is robust to the violation of HWE.
- The additive model is a general model for the association test and conserved for other modes of inheritance.
- The logistic regression model can be used for the association.

$$\log \frac{P(Y=1|X)}{P(Y=0|X)} = \beta_0 + \beta_1 X$$

  where $Y = 1$ for a case and 0 for a control, X=count of allele a in the genotype.
- Testing $H_0 : \beta_1 = 0$ against $H_a : \beta_2 \neq 0$ is to test the association between a SNP and the disease/trait.

Spurious association: An Example

- Consider Cystic Fibrosis, a rare disease that is only present in Caucasian.
- Let a SNP has two alleles: 1 and 2, with frequencies of allele 1, $p_C = 0.6$ in Caucasian population, and $p_A = 0.2$ in Asian population. The SNP is not associate to the disease locus.
- Study design: random samples of cases and controls from Scarborough, Ontario. In Scarborough, population size of Caucasians and Asians are approximately equal.
- Take random sample of $n_1 = 100$ cases from the affected population, namely, the affected individuals in Scarborough. Remember, disease only occurs in Caucasian population. Thus, we assume all the cases are Caucasians.
- Take random sample of $n_2 = 100$ from the unaffected population in Scarborough. We expected to have about 50% and 50% Caucasians and Asians in control group.

Spurious Association: Continue to the example

- Data we collected:

	SNP		
group	allele 1	allele 2	Total
case	118	82	$2n_1$=200
control	78	122	$2n_2$=200
	200	200	400

- Let X_1 and X_2 be the number of allele 1 in case group and control group respectively. so, we have $X_1 = 118$, $X_2 = 78$ in this example.
- In cases, since all individuals are Caucasians, the expected number of the allele 1 is $2n_1p_c = 2 \times 100 \times 0.6 = 120$.

  In control, since it is about 50% and 50% of Caucasian and Asians, the expected frequency of allele 1 is $\frac{1}{2}p_c + \frac{1}{2}p_a = 0.4$. Thus, the expected number of allele 1 is $2 \times 100 \times 0.4 = 80$.

Spurious Association: Continue to the example

- Suppose we didn't have the above population stratification information in our data. We perform a case-control association test.
  - Let p_0 and p_1 be the frequencies of allele 1 in the control group and case group.
  - If the marker is not associated with disease, the allelic distribution will be the same between the case and control population.
  - We test $H_0 : p_0 = p_1$ against $H_a : p_0 \neq p_1$.
  - We can use the Pearson χ^2 test to test the association.

Peason χ^2 test in R

We can perform the Pearson χ^2 test in R.

```
> x=matrix(c(118, 78, 82, 122), ncol=2)
> x
     [,1] [,2]
[1,]  118   82
[2,]   78  122
> chisq.test(x)

Pearson's Chi-squared test with Yates' continuity
correction

data:  x
X-squared = 15.2161, df = 1, p-value = 9.588e-05

>chisq.test(x)$p.value

[1] 9.588316e-05
```

Result of Spurious Association

- As p-value is so small, we reject the H_0.
- We conclude that there is strong and significant evidence to suggest that the case and control group do not have the same allelic distribution, and therefor the SNPs we test is associated with the disease with strong and significant evidence.

  However, the observed association is spurious association due to the population stratification, not the allelic association with the disease locus due to the LD between the two loci.
- Thus, for population based association studies, matching is an important step for a study design in order to eliminate the spurious association. Or a pre-study is necessarily for identifying the spurious association.

Contol of Spurious Association

- Better study design? Try to match the cases and controls by age, sex, ethnicity, environmental exposures, etc.

  However, it is difficult to sample well match case and control , especially, it is difficult to recruit appropriate control matched for ethnicity in admixed population such as New York.
- Can we recruit controls from relatives of cases?
  - Cases and controls are not independent, as a case and control come from the same family may receive the same allele from their common ancestor.
  - The dependence (genetically) of case and control must be taken into account in the analysis.
    - Bourgain et al. (2005). Novel case-control test in a founder population identifies P-selectin as an atopy susceptibility locus (2003) American Journal of Human Genetics 73:612-626.
    - Feng et al. (2011). Generalized genetic association study with samples of related individuals. Annals of Applied Statistics 5(3): 2109-2130.

Contol of Spurious Association

- Genomic control methods use all available markers data throughout the genome to detect potential spurious association.

  Basic principle: population stratification should affect all marker loci similarly, so that any marker showing an association with the disease significantly above the average (over the all the markers throughout the genome) is suspected of being linked to the disease locus.

  Delvin B, Roeder K, Wasserman L (2001). Genomic control, a new approach to genetic-based association studies. Theoretical Population Biology, 60:155-166.

Control of Spurious Association

- Structured association methods based on all available markers throughout the genome to detect population structure.

  Basic principle: infer the population structure (identify subgroups in a given population) from the all markers, and then take into account the inferred structure in the testing procedure.

  Pitchard JK, Donnelly PJ (2001). Case-control studies of association in structured or admixed populations. Theoretical Population Biology, 60;227-237.

Association Analysis with Quantitative Trait

- Quantitative trait: the trait is a continuous measure/
- The quantitative trait is usually assumed to be normally distribution.
- A SNP is considered as a covariate.
- A simple linear regression model can be used to link the SNP to quantitative trait.
- Suppose there is a sample of size n subjects.
  - Let Y_i be the measure of quantitative trait of the ith subject.
  - Let X_i be the genotype of the SNP in the dosage format that $X_i = 0, 1, 2$ representing the count of minor allele.
  - $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \quad i = 1, ..., n, \quad \varepsilon_i \overset{iid}{\sim} N(0, \sigma^2),$

$$Y_i | X_i \sim N(\beta_0 + \beta_1 X_i, \sigma^2),$$

That is, the mean of Y_i depending on the value of X_i.

Association Analysis with Quantitative Trait

- If a SNP is not associated with any underlying QTL, the coefficient associate with X_i is zero, i.e $\beta_1 = 0$.
- We can perform a statistical hypothesis test for the association:

$$H_0 : \beta_1 = 0, \quad \text{agains} \quad H_a : \beta_1 \neq 0$$

We use the t statistic:

$$t = \frac{\hat{\beta}_1}{SE(\hat{\beta}_1)} \sim t_{n-2}, \quad \text{under the } H_0,$$

Association Analysis with Quantitative Trait

- If you want to account for other covariates effects such as gender, age, BMI, high density lipoprotein (HDL), .., on the blood pressure, you can can used a multiple linear regression model, and the SNP is one of the covariates. For example
  Y: blood pressure
  X_1: SNP
  X_2: age
  X_3: gender, 1 for female, and 0 for male
  X_4: BMI
  X_5: HDL

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + ... + \beta_5 X_{i5} + \varepsilon_i, \quad i = 1, ..., n, \quad \varepsilon_i \overset{iid}{\sim} N(0, \sigma^2)$$

$$Y_i | X_i \sim N(\beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + ... + \beta_5 X_{i5}, \sigma^2),$$

That is, the mean of Y_i depending on the values of $X_{i1}, ..., X_{i5}$.

Association Analysis with Quantitative Trait

- We can perform a statistical hypothesis test for the association between the SNP and the trait (BP) here

$$H_0 : \beta_1 = 0, \quad \text{agains} \quad H_a : \beta_1 \neq 0$$

We use the t statistic:

$$t = \frac{\hat{\beta}_1}{SE(\hat{\beta}_1)} \sim t_{n-6}, \quad \text{under the } H_0,$$

- This association test can be done by using the R "lm( )" function.

Association Analysis with Quantitative Trait

- Again, when testing the genetic association with a quantitative trait, we need to be careful about
  - the population stratification,
  - whether all subjects in the sample are independent? are there correlated subjects?
  - are there admixture subjects,
  - ...

Genome-wide Association Studies (GWAS)

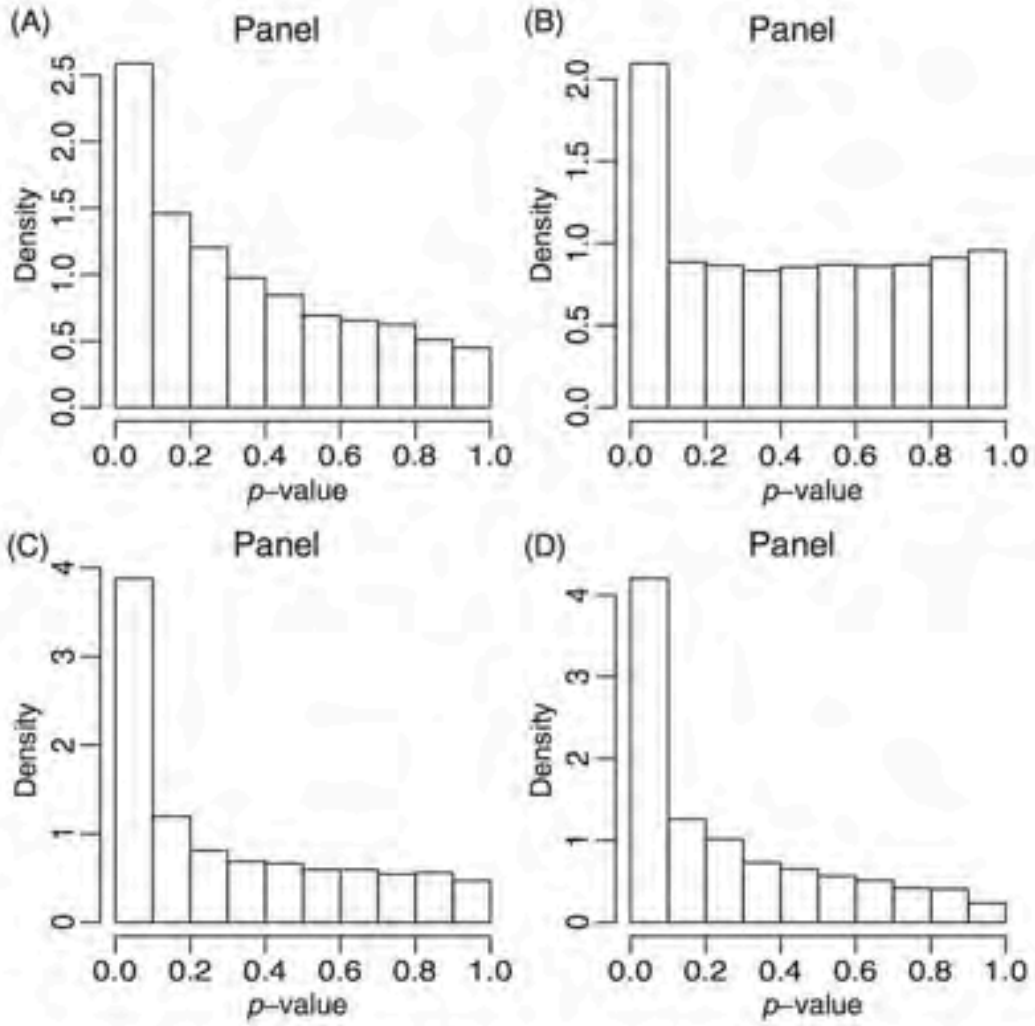
- GWAS refers to the screening of a genome-wide set of genetic variants (usually SNPs) for variant(s) or genome region(s) that are associated with the trait(s) of interested.
- The number of SNPs determines the number of hypothesis tests.
  - A coverage of 100,000 SNPs means performing 100,000 hypothesis tests.
  - 10,000 tests mean 100,000 p-values for the significance level of the association between each SNP and the trait.
- Issue of controlling the overall error rate among a large number of test.
  - Family-wise error rate (FWER): $\alpha_F = 1 - (1 - \alpha)^K$, $K =$ number of SNPs, $\alpha_F =$ FWER.
  - Bonferroni control: $\alpha = \alpha_F / K$, $i = 1, ..., K$.
  - False discover rate (FDR).

Genome-wide Association Studies (GWAS)

- It is always good to take a look at the histogram of p-values.
  - Under H_0 that there is no association between a SNP and the trait, p-value $\sim$ Unif[0, 1] distribution.
  - Among tens or hundreds thousands SNPs, majority are not associated with the trait, only a small proportion of them are associated with the trait.
  - Majority of the p-values follow a Unif[0, 1] distribution.
- In a GWAS of a million or hundred SNPs, a cut-off of $10^{-}5$ of p-values is often used. That is, the SNPs that has p-values less than 0.00001 will be further investigated.
- The overall p-value cut-off is subject to the study.
- For GWAS, we sometime use a Mahattan plot for the illustration of the significant SNPs.

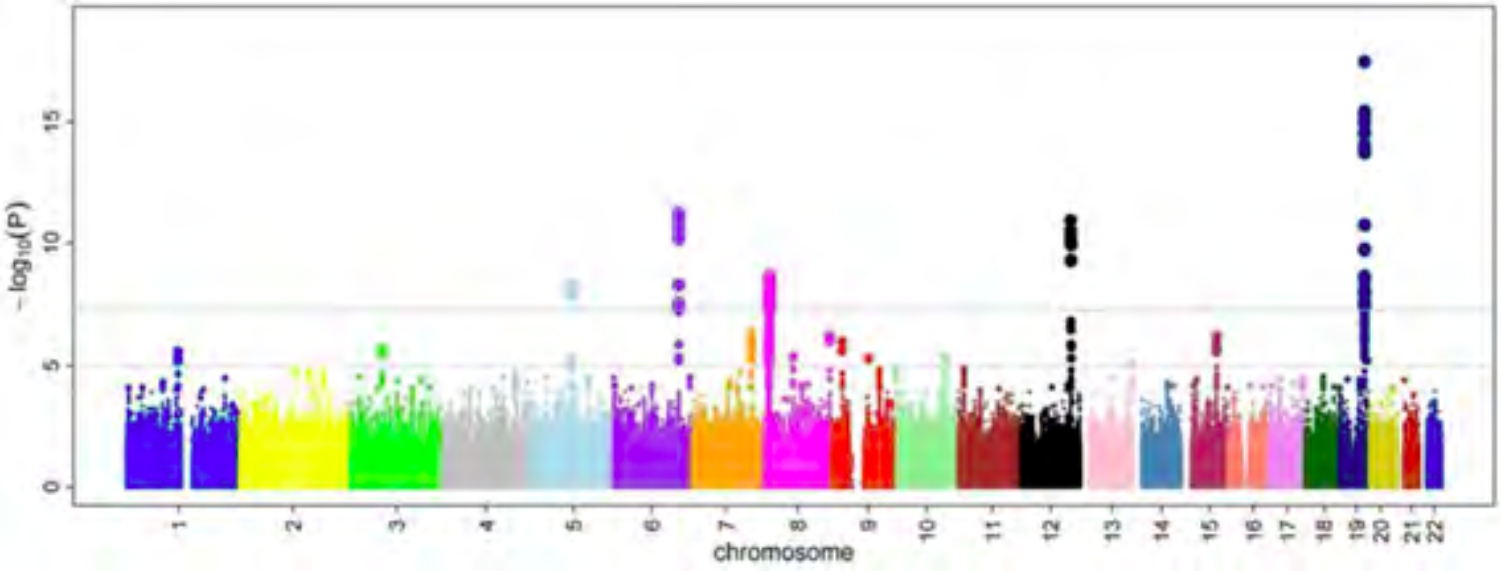
P-value Histogram

Histograms of p-values from four studies of microarray gene expression for identifying genes differentially expressed under different conditions.



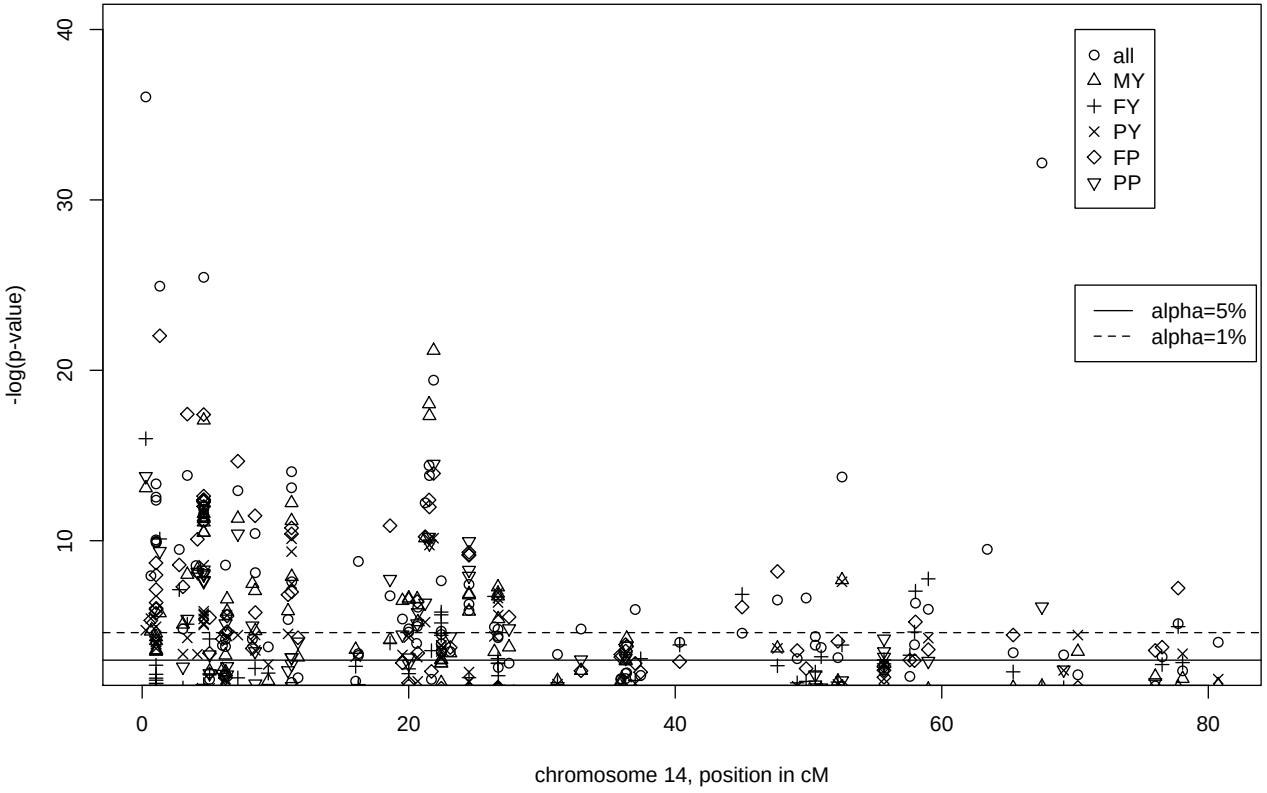
Manhattan plot

An example of Mahattan Plot of a GWAs with a single trait.



Manhattan plot

An example of association tests of SNPs on Chromosome 14 of a GWAS with mutliple economic traits of Holstein dairy cattle.



General procedure for a genetic study

- Study design: observational study, experimental study (non-human), clinical study (human), population-based, family based, isolated/founder-based population, longitudinal study,....
- Variables in the data set:
  - Outcome of interest: response variable(s), trait(s), disease affection status or stages.
  - Covariates, explanatory variables:
  - Genetic variables: commonly SNPs, gene-based, genome-wide coverage (GWAs).
  - Metadata: age, sex, lifestyle covariate such smoking status, alcohol consumption, weight, and BMI, treatments (of disease), ...
- Objective: identify common factors that contribute to the variation of the response, or, disease risk.

General procedure for a genetic study

- Environmental risk factor: smoking status, alcoholic assumptions, different type of fertilizer to the crops, soil type, feeding materials, ...
- Genetic factors:
  - major gene effects
  - polygenic effects (multiple genes, and each with a small contribution)
  - genetic pleiotropic effects
  - gene-gene interaction
  - gene-environmental interaction
  - time-varying gene effects
  - gene-treatment interactions (personalized medicine)

General procedure for a genetic study

- GWAs are usually the first step to screen relevant SNPs, different for different study designs.
- After GWA screening, more complex or sophisticated statistical models are considered. For example:
  - LASSO-type of variable selection methods to build a complex model based on the significant SNPs after GWA screening.
  - Genomic best linear unbiased prediction (GBLUP) model: predicting genomic breeding value (GEBV), based on mixed effects models, the relationship matrix among animals is computed based on the genotype rather than the genealogy.
- Haplotype association analysis: gene-based, genome-wide
  - Haplotype inference: infer haplotype forms based on genetoype data.
  - Estimate haplotypes effects and test for the significance of these effects.
- Missing SNPs imputation: based on LD structure among neighboring SNPs.